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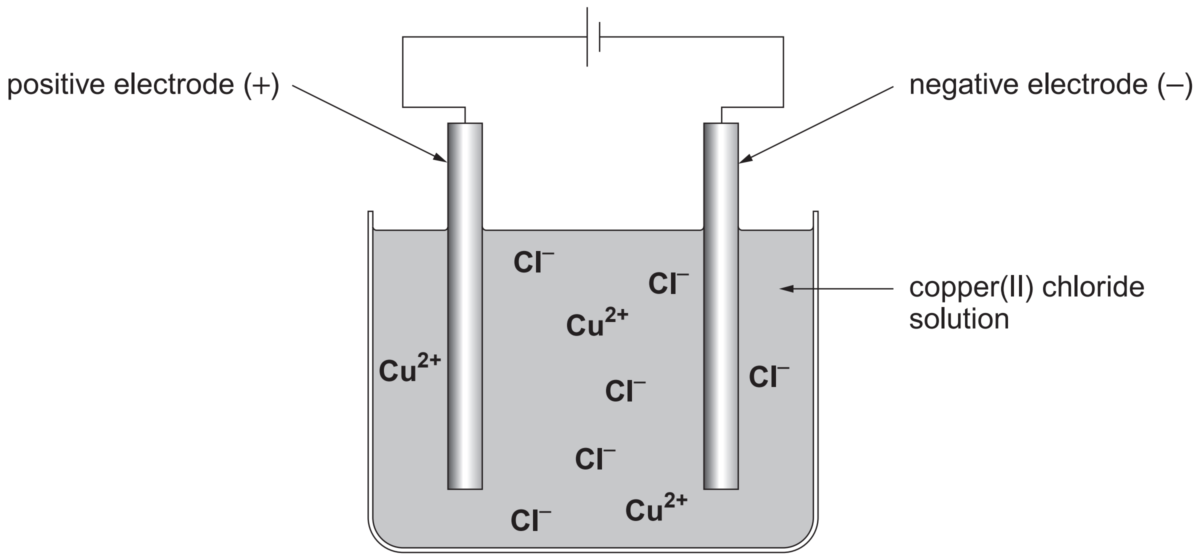
2. (a) Corey and Bailey used the following apparatus to obtain copper from copper(II) chloride solution.



(i)

electrolysis	**copper**	**anode**
cathode	**chlorine**	**displacement**
copper(II) chloride	**neutralisation**	

Choose words from the box to complete the following sentences. [3]

The electrolyte used in this process is ....

Copper is formed at the negative electrode. This electrode is called the

....

The breaking down of a compound using an electric current is called

....

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(ii) Copper(II) chloride contains the ions Cu^{2+} and Cl^{-}.

Give the formula of copper(II) chloride. [1]

....

(iii) During the process, copper ions become copper atoms by gaining electrons.

Tick (✓) the box next to the equation that shows this. [1]

$Cu - 2e^{-} \longrightarrow Cu^{2+}$	☐
$Cu + 2e^{-} \longrightarrow Cu^{2+}$	☐
$Cu^{2+} + 2e^{-} \longrightarrow Cu$	☐
$Cu^{2+} - 2e^{-} \longrightarrow Cu$	☐

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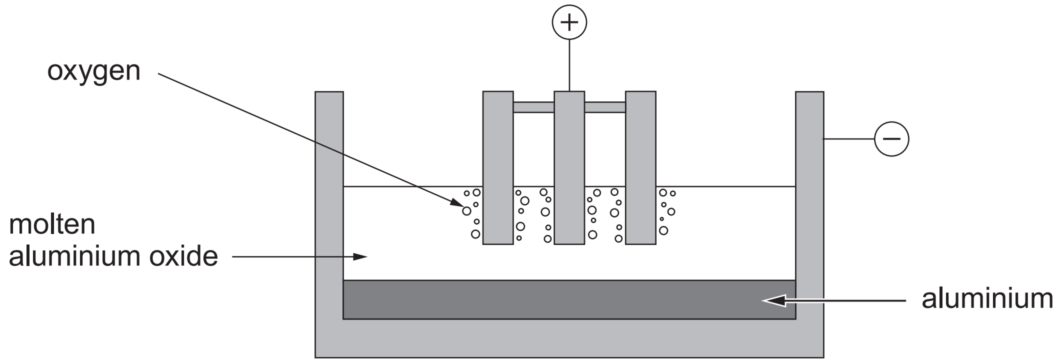
(b) The diagram represents the industrial extraction of aluminium from its ore.



(i) Give the reason why the aluminium oxide must be molten during the extraction.

Tick (✓) the correct box. [1]

to release the oxygen gas from the aluminium oxide ☐

to allow the aluminium ions and oxide ions to move ☐

to allow the aluminium to leave the cell ☐

to speed up the process ☐

(ii) Put a number in the box to balance the equation that represents the overall reaction taking place. [1]

$$2\,Al_2O_3 \longrightarrow \boxed{}\,Al + 3\,O_2$$

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(iii) Tick (✓) the boxes next to the **two** factors that are important when choosing the location for aluminium plants in the UK. [2]

Aluminium plants in the UK should be close to

landfill sites	☐
power stations	☐
limestone quarries	☐
oil refineries	☐
coastal ports	☐
coal mines	☐

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(c) Aluminium is commonly used to make overhead power cables.

Other than being a good electrical conductor, give **two** properties which make aluminium a suitable material for making overhead power cables. [2]

Property 1 ....

Property 2 ....

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